



Direct-path Relative Harmonic Coefficients Detection for Multi-source Direction-of-Arrival Estimation in Reverberant Environments

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Abstract

The identification of direct-path recordings for direction-of-arrival (DOA) estimation has been recognized as a crucial factor in enhancing the localization robustness under reverberant environments, especially for the challenging cases involving multiple acoustic sources. In particular, the recently introduced spatial feature denoted *relative harmonic coefficients (RHC)* has been extensively investigated for DOA estimation due to a strong association with spatial cues. Building upon this, this paper presents a multi-source localization method that utilizes higher-order RHC vector to detect direct-path dominant (DPD) bins in the time-frequency (TF) domain. Specifically, DPD bins are effectively identified by applying an orthogonal projection matrix (OPM) to the RHC vector. Moreover, we integrate the OPM with a closed-form single source DOA estimator, thus circumventing the exhaustive search over a dense spatial grid. Extensive experimental evaluations using both simulated and real-world recordings confirm the proposed algorithm's effectiveness, and superiority over baseline methods.

Index Terms: direction-of-arrival estimation, multi-source, reverberation, spherical microphone arrays

1. Introduction

Acoustic source direction-of-arrival (DOA) estimation continues to be a prominent topic in the audio signal processing community, with the goal of recovering the directional information of source(s). Accurate knowledge of source positions contributes to a range of practical audio-only applications, such as robot navigation, cochlear implants and intelligent surveillance systems [1]. Furthermore, numerous acoustic signal processing techniques, including but not limited to dereverberation [2], beamforming [3], and spatial acoustic rendering [4], mostly require the *prior* knowledge of DOA to enhance their performance or achieve the intended objectives.

Recently, spherical microphone arrays (SMA) have gained increasing popularity for DOA estimation due to several unique advantages: (i) providing comprehensive spatial cues over a two-dimensional (2-D) directional space; and (ii) decomposing into the spherical harmonics (SH) domain, enabling the decoupling of frequency- and angular-dependent components [5]. This facilitates significant advancements in source localization algorithms with such arrays, including beamforming [6], subspace [7, 8], pseudo-intensity vector (PIV) [9, 10], and relative harmonic coefficient (RHC)-based approaches [11, 12]. However, in practical acoustic environments, acoustic reflections due to reverberation pose a major challenge as they severely distort the propagation of direct-path recordings, thus significantly degrades localization performance in multi-source scenarios. One of the most efficient solution is to detect the direct-path dom-

inant (DPD) components prior to source localization, which leverages the sparsity of speech signals in the time-frequency (TF) domain. Consequently, considerable efforts have been devoted to developing reliable DPD bin detectors, to enhancing the robustness of acoustic localization approaches against reverberations [13, 14, 15].

One early method for detecting DPD components checks whether the rank of the correlation matrix is unity with a coherence test [16]. Later, Nadiri et al. proposed identifying DPD bins by comparing the ratio of the largest- and second eigenvalues of spatial covariance matrix, where frequency smoothing techniques are applied to de-correlate the mixed signals [13]. In another study [17], sound field directivity was utilized to guide the extraction of DPD bins, achieving significant performance improvements. Instead of focusing on bin-level detection, the authors in [18] identified DPD TF zones by calculating the correlation coefficient of magnitude between any two sensors. Recently, Hu et al. introduced a promising SH-domain feature, known as the RHC, for DOA estimation, which serves as the SH-domain counterpart to the relative transfer function (RTF). Two key properties of RHC are summarized as follows: (i) it is independent of the source signal, and (ii) it depends solely on the source's DOA. Both advantages have facilitated the development of 2-D grid search [19], decoupled [20], and closed-form multi-source DOA estimation algorithms [21]. However, grid-search based approaches suffer from high computational complexity, while the closed-form method not fully utilizes the higher-order SH coefficients. As a result, existing RHC-based methods remain suboptimal in practice.

Motivated by the advantages of RHC, this paper proposes a novel direct-path components detection method by introducing an orthogonal projection matrix (OPM) to the higher-order RHC vector. It is shown that the l_2 norm of the vector, obtained by applying an properly designed OPM to direct-path RHC, become zero. This unique property can effectively guide the identification of DPD bin in the TF domain. To improve computational efficiency, the establishment of OPM is skillfully associated with a closed-form single-source DOA estimator, eliminating the need for tedious grid search procedures.

2. System Model

2.1. Spatial Domain Model

Consider a higher-order SMA consisting of J omni-directional sensors to record Q concurrent sources in noisy and reverberant scenarios. With the short-time Fourier transform (STFT), the sound pressure measured by the j -th sensor is modeled as,

$$X_j(l, k) = \sum_{q=1}^Q S_q(l, k) H_{j,q}(l, k) + V_j(l, k), \quad (1)$$

the time frame and frequency are indexed by l and k , respectively, $X_j(l, k)$ is the sound pressure captured by the j -th ($j = 1, 2, \dots, J$) microphone, $S(l, k)$ denotes the q -th incident signal, $V_j(l, k)$ denotes the noise signal at the j -th channel, $H_{j,q}(l, k)$ denotes the acoustic transfer function (ATF) from the q -th source to the j -th microphone. The noise signal is assumed to be uncorrelated with the source signal in the sense of statistics. Assuming far-field hypothesis, the ATF in multi-path scenarios is expressed by [22],

$$H_{j,q}(l, k) = G_q^{(d)} e^{ikr\hat{\mathbf{g}}_q^T \mathbf{x}_j} + \int_{\hat{\mathbf{g}}} G_q^{(r)} e^{ikr\hat{\mathbf{g}}^T \mathbf{x}_j} d\hat{\mathbf{g}}, \quad (2)$$

where $G_q^{(d)}$ denotes the direct-path gain of the q -th source that is uniformly set to 1 for simplicity, $G_q^{(r)}$ denotes the reflection gain along the direction of $\hat{\mathbf{g}}$ for the q -th source, $\hat{\mathbf{g}}_q = [\sin\vartheta_q \cos\varphi_q, \sin\vartheta_q \sin\varphi_q, \cos\vartheta_q]^T$ denotes the unit vector of the q -th source location, ϑ_q and φ_q are the respective elevation and azimuth with correspond range $[0, \pi]$ and $(0, 2\pi]$, $\mathbf{x}_j$ is defined in a similar way to $\hat{\mathbf{g}}_q$, but it denotes the position of the j -th microphone. i is the imaginary unit, r is the array radius, $k = 2\pi f/c$ is the wave number with the frequency f and the sound speed c . For the sake of simplicity, the dependence on the time index l is omitted in subsequent chapters.

2.2. Spherical Harmonic Decomposition

Multi-channel measurements acquired by SMAs are typically decomposed into the SH domain due to the decoupling characteristics of frequency- and angular-dependent components, making DOA estimation more attractive. Hence, (1) is able to be represented as [23],

$$X_j(k) = \sum_{n=0}^N \sum_{m=-n}^n \alpha_{nm}(k) b_n(kr) Y_{nm}(\Psi_j), \quad (3)$$

where $N = \lceil kr \rceil$ is the maximum truncated SH order due to the high-pass nature of the spherical Bessel functions [24], and $\lceil \cdot \rceil$ denotes the ceiling operator. Notably, the sound field order is closely relate to the number of sensors, i.e., $J \geq (N+1)^2$ must be satisfied to prevent spatial aliasing. $b_n(kr)$ is the radial function and various types of sphere has different expression [25],

$$b_n(kr) = \begin{cases} i^n j_n(kr), & \text{for open sphere} \\ i^n \left[j_n(kr) - \frac{j'_n(kr)}{h'_n(kr)} h_n(kr) \right], & \text{for rigid sphere} \end{cases} \quad (4)$$

where $(\cdot)'$ denotes the first derivative of a function, $j_n(\cdot)$ and $h_n(\cdot)$ are the respective first kind of spherical Bessel and second kind of spherical Hankel functions. $Y_{nm}(\Psi)$ is the SH function of order n and degree m with 2-D space orthogonality,

$$Y_{nm}(\Psi) = \sqrt{\frac{(2n+1)(n-m)!}{4\pi(n+m)!}} P_{nm}(\cos\theta) e^{im\phi}, \quad (5)$$

where $P_{nm}(\cdot)$ is the associated Legendre polynomial, $(\cdot)!$ denotes the factorial operator. The SH coefficients $\alpha_{nm}(k)$ characterize the sound field information within the interior region. Rewrite all the terms in matrix form for clarity, multi-channel observations is capable of expressed as,

$$\begin{aligned} \mathbf{x}(k) &= [X_1(k) \ X_2(k) \ \dots \ X_J(k)]^T \\ &= \mathbf{T}(k) \boldsymbol{\alpha}_N(k) = \mathbf{V} \mathbf{B}(k) \boldsymbol{\alpha}_N(k), \end{aligned} \quad (6)$$

where

$$\begin{aligned} \mathbf{V} &= \begin{bmatrix} Y_{00}(\Psi_1) & \dots & Y_{NN}(\Psi_1) \\ \vdots & \ddots & \vdots \\ Y_{00}(\Psi_J) & \dots & Y_{NN}(\Psi_J) \end{bmatrix} \\ \mathbf{B}(k) &= \text{diag}(b_0(kr), b_1(kr), \dots, b_N(kr)) \\ \boldsymbol{\alpha}_N(k) &= [\alpha_{00}(k), \alpha_{1,-1}(k), \dots, \alpha_{NN}(k)]^T \end{aligned} \quad (7)$$

Since $\mathbf{T}(k)$ denotes a $J \times (N+1)^2$ matrix, whereby $(N+1)^2 < J$, thus the equation (6) is over-determined. By solving the least squares problem for the over-determined system, the SH coefficient vector $\boldsymbol{\alpha}_N(k)$ can be estimated as [26],

$$\boldsymbol{\alpha}_N(k) = \mathbf{B}^{-1}(k) (\mathbf{V}^T \mathbf{V})^{-1} \mathbf{V}^T \mathbf{x}(k). \quad (8)$$

3. Proposed Method

3.1. Orthogonal Projection on RHC Vector

The RHC is defined as the ratio of arbitrary-order SH coefficients to that of zeroth-order [12], and its vector form is,

$$\boldsymbol{\beta}_N(k) = \frac{\boldsymbol{\alpha}_N(k)}{\mathbf{e}_1 \boldsymbol{\alpha}_N(k)} = [1, \beta_{1,-1}(k), \dots, \beta_{NN}(k)]^T, \quad (9)$$

where $\mathbf{e}_1 = [1, 0, \dots, 0]$ with a length of $(N+1)^2$. According to free space Green's function by considering a TF bin contains significant contribution from the q -th source and neglecting the noise term, the analytic expression of the (n, m) -th SH coefficient, $\alpha_{nm}(k)$, is given by [11],

$$\alpha_{nm}(k) = S_q(k) 4\pi Y_{nm}^*(\Phi_q), \quad (10)$$

where $(\cdot)^*$ denotes the conjugate operator, $\Phi_q = (\vartheta_q, \varphi_q)$ is the DOA of the q -th source. Substitute (10) into (9) and remove the constant entry '1', direct-path RHC vector is expressed as,

$$\boldsymbol{\beta}_N(\Phi_q) = [2\sqrt{\pi} Y_{1,-1}^*(\Phi_q), \dots, 2\sqrt{\pi} Y_{NN}^*(\Phi_q)]^T, \quad (11)$$

From (11), one can see that the frequency independence of RHC arises due to the cancellation of source contributions, leaving only the source DOA as the only degree of freedom, thus we replace $\boldsymbol{\beta}_N(k)$ with $\boldsymbol{\beta}_N(\Phi_q)$. Therefore, the OMP associated with N -order direct-path RHC vector is defined as,

$$\mathbf{P}_N(\Phi) = \mathbf{I}_N - \frac{\boldsymbol{\beta}_N(\Phi) \boldsymbol{\beta}_N^H(\Phi)}{\|\boldsymbol{\beta}_N(\Phi)\|^2}, \quad (12)$$

where $\mathbf{I}_N$ is an identity matrix, $\|\cdot\|$ denotes an l_2 norm, and $\boldsymbol{\beta}_N(\Phi)$ is a pre-defined vector that depends on only the directional information. Apply the OPM to the (11), the DOA of the q -th source can be determined by searching for the minimum value of the l_2 norm over a continuous 2-D grid, which is an efficient narrow-band DOA estimator,

$$\Phi_q = \arg \min_{\Phi} \|\mathbf{P}_N(\Phi) \boldsymbol{\beta}_N(\Phi_q)\|. \quad (13)$$

We emphasize that the l_2 norm of the transformed vector tends to be zero-valued, i.e., $\|\mathbf{P}_N(\Phi_q) \boldsymbol{\beta}_N(\Phi_q)\| = 0$, which forms the foundation of the proposed DPD component detection.

It has been noted that room reflections in multi-source reverberant scenarios lead to the presence of a lot of multi-path overlapped (MPO) bins, and the SH coefficient of such bin is represented by a linear superposition of multiple plane waves,

$$\alpha_{nm}(k) = 4\pi \sum_{q=1}^{Q'} [S_q(k) Y_{nm}^*(\Phi_q) + \int_{\hat{\mathbf{g}}} S_q^{(\hat{\mathbf{g}})}(k) Y_{nm}^*(\Phi_{\hat{\mathbf{g}}}) d\hat{\mathbf{g}}]. \quad (14)$$

The analytic expression of the (n, m) -th element in the RHC vector $\beta_N(k)$ for MPO bin is,

$$\begin{aligned}\beta_{nm}(k) &= \frac{4\pi \sum_{q=1}^{Q'} \left[S_q(k) Y_{nm}^*(\Phi_q) + \int_{\hat{\mathbf{y}}} S_q^{\hat{\mathbf{y}}}(k) Y_{nm}^*(\Phi_{\hat{\mathbf{y}}}) \right]}{\sum_{q=1}^{Q'} \left[S_q(k) Y_{00}^*(\Phi_q) + \int_{\hat{\mathbf{y}}} S_q^{\hat{\mathbf{y}}}(k) Y_{00}^*(\Phi_{\hat{\mathbf{y}}}) \right]} \\ &= 2\sqrt{\pi} \sum_{q=1}^{Q'} \left[\bar{S}_q(k) Y_{nm}^*(\Phi_q) + \int_{\hat{\mathbf{y}}} \bar{S}_q^{\hat{\mathbf{y}}}(k) Y_{nm}^*(\Phi_{\hat{\mathbf{y}}}) \right]\end{aligned}\quad (15)$$

where

$$\bar{S}_q(k) = \frac{S_q(k)}{\sum_{q=1}^{Q'} \left[S_q(k) + \int_{\hat{\mathbf{y}}} S_q^{\hat{\mathbf{y}}}(k) \right]}, \quad (16)$$

and $\bar{S}_q^{\hat{\mathbf{y}}}(k)$ is defined in a similar way. We see that the OPM defined in (12) theoretically fails to achieve an effective orthogonal transformation for MOP bins. Additionally, we consider a type of TF bin composed solely of noise and/or late reverberation components, referred to source-absent (SA) bins. These bins can be treated as random complex numbers due to lack of explicit analytical expression, making it impossible to properly transform them using the defined OPM. Overall, both types of non-DPD bins behave differently from the direct-path RHC, highlighting the efficacy of leveraging the OPM for direct-path component detection. As mentioned above, tedious 2-D grid search results in significant computational overhead, motivating the adoption of a closed-form single-source DOA estimator to address this issue (see next subsection). Specifically, the instantaneous DOA for each TF bin is pre-calculated to directly determine the construction of the OPM, after which the l_2 norm of the transformed vector is utilized to identify DPD bins.

3.2. Direct-path RHC Detection

Motivated by the distinction between DPD and non-DPD bins from the perspective of orthogonal projection on RHC vector, we define the following detection rule to identify direct-path RHC,

$$D(k) = \frac{1}{e^{||\mathbf{P}_N(\Phi)\tilde{\beta}_N(k)||}}, \quad (17)$$

and the DPD bin set is given by,

$$\mathcal{O}_{\text{DPD}} = \{(l, k) | D(l, k) > \gamma\}, \quad (18)$$

where $1 \leq k \leq K, 1 \leq l \leq L, \gamma \in [0, 1]$ is a pre-defined threshold for identifying DPD bin, and $\tilde{\beta}_N(k)$ denotes the estimated RHC vector, as introduced below.

To enhance the robustness of algorithm in noisy conditions, RHC is suggested to be estimated by calculating the ratio of cross power spectral density (PSD) to self-PSD due to the speech stationarity over short time frames [20]. The (n, m) -th element in the estimated RHC vector is obtained by,

$$\tilde{\beta}_{nm}(k) = \frac{\mathbb{E}[\alpha_{nm}(k)\alpha_{00}^*(k)]}{\mathbb{E}[\alpha_{00}(k)\alpha_{00}^*(k)]}, \quad (19)$$

where $\mathbb{E}[\cdot]$ denotes the statistical expectation operator, and

$$\mathbb{E}[\alpha_{nm}(k)\alpha_{00}^*(k)] = \frac{1}{T} \sum_l \alpha_{nm}(l, k)\alpha_{00}^*(l, k), \quad (20)$$

$$\mathbb{E}[\alpha_{00}(k)\alpha_{00}^*(k)] = \frac{1}{T} \sum_l \alpha_{00}(l, k)\alpha_{00}^*(l, k). \quad (21)$$

3.3. DOA Estimation

In this work, the use of a closed-form single-source DOA estimator would significantly improve computational efficiency, since both the identification of direct-path RHC and its corresponding instantaneous DOA estimates are conducted simultaneously. Inspired by the method developed in [11], the direction vector is derived by multiplying a complex-to-real transformation matrix to the first-order RHC vector, i.e.,

$$\mathbf{M}\tilde{\beta}_1(k) = \begin{bmatrix} \lambda_{1,-1}(k) \\ \lambda_{1,0}(k) \\ \lambda_{1,1}(k) \end{bmatrix} = \begin{bmatrix} \sin\vartheta\sin\varphi \\ \cos\vartheta \\ \sin\vartheta\cos\varphi \end{bmatrix}, \quad (22)$$

where

$$\mathbf{M} = \frac{1}{2\sqrt{2}\pi} \begin{bmatrix} -i & 0 & -i \\ 0 & \sqrt{2} & 0 \\ 1 & 0 & -1 \end{bmatrix}. \quad (23)$$

Consequently, the respective azimuth and elevation at one TF bin can be directly extracted, such as $\varphi(k) = \arctan\{\mathcal{R}[\lambda_{1,-1}(k)]/\mathcal{R}[\lambda_{1,1}(k)]\}$ and $\vartheta(k) = \arctan\left\{\sqrt{\mathcal{R}[\lambda_{1,-1}(k)]^2 + \mathcal{R}[\lambda_{1,1}(k)]^2}/\mathcal{R}[\lambda_{1,0}(k)]\right\}$,

where $\mathcal{R}[\cdot]$ denotes the real operator. Implement the proposed method over the entire STFT domain, where the DOA set for the detected DPD bins is represented as Ω_{DPD} . Mapping out this set yields multiple distinct peaks, corresponding to the DOA estimates of individual source. To enhance localization performance, we adopt the kernel density estimation (KDE) algorithm to further smooth the histogram for more accurate results. The reader is referred to [27, 28] for more details.

4. Experiments

This section evaluates the proposed algorithm using both the simulated and real-world recordings. Moreover, we present the comparison results with four baseline approaches, i.e., WDO (using all STFT bins within the given frequency band), Closed-form [21], Decoupled [20], and DPD-MUSIC methods [13]. Localization accuracy is measured by the commonly used metric of mean absolute estimated error (MAEE).

4.1. Simulated Environments

We simulate a rectangular room with dimensions of $6\text{m} \times 4\text{m} \times 3\text{m}$ using the image source method [29]. An open spherical array equipped with 32 capsules is positioned at the center of the room to capture the interior sound field. All source signals are assumed to be static and placed 1m away from the array origin. Dry speech signals are randomly selected from the TIMIT corpus at a sampling rate of 16 kHz [30] and are convolved with the room impulse response (RIR). Gaussian white noise with an SNR level of 15 dB is added to each channel to generate noisy measurements, which are then segmented into 2-second signal blocks for DOA estimation. The noisy signals are transformed into the frequency domain using the STFT with a Hamming window of 512 samples, a 50% overlap between frames, and a 2048-sample fast Fourier transform (FFT). For DOA estimation, a limited frequency range of [2600, 3600] Hz is utilized, as SH coefficients can be estimated up to the 3rd order avoiding the "Bessel-zero" problem [31].

Before evaluating the localization accuracy, we first examine the capability of the proposed DPD bin detection method by simulating three plane waves propagating from $(48^\circ, 112^\circ)$, $(89^\circ, 76^\circ)$ and $(217^\circ, 142^\circ)$, and they are polluted by 15 dB

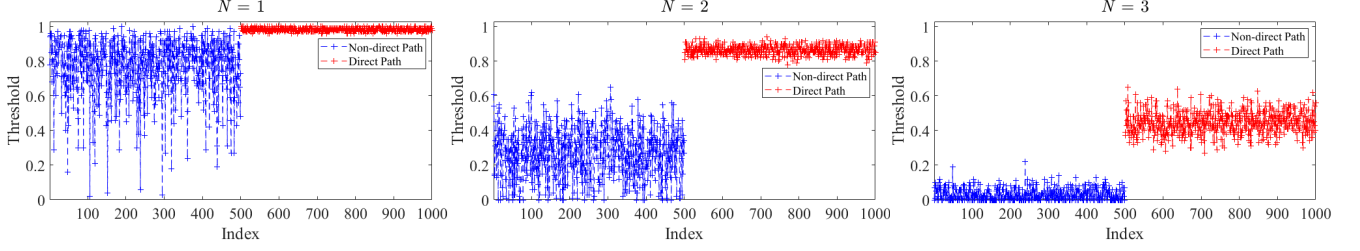


Figure 1: Direct-path detection using simulated plane waves for various sound field orders ($N = 1, 2, 3$).

Table 1: Results of the comparison methods in the reverberant scenarios with different T_{60} s and source number

Method	2 sources						3 sources					
MAEE	0.2s	0.35s	0.5s	0.65s	0.8s	avg.	0.2s	0.35s	0.5s	0.65s	0.8s	avg.
WDO	3.69°	4.21°	4.36°	4.74°	5.43°	4.49°	4.32°	4.83°	4.96°	5.08°	5.50°	4.90°
Closed-form	2.21°	2.81°	3.39°	3.36°	3.77°	3.11°	2.63°	3.61°	3.59°	4.52°	4.67°	3.81°
Decoupled	2.24°	2.52°	2.58°	2.75°	2.85°	2.59°	2.44°	2.73°	3.11°	3.67°	3.72°	3.13°
DPD-MUSIC	1.81°	2.45°	2.82°	2.93°	3.28°	2.66°	1.75°	2.34°	2.93°	2.98°	3.61°	2.72°
Proposed	1.98°	2.22°	2.63°	2.69°	3.04°	2.51°	1.82°	2.69°	2.78°	2.94°	3.33°	2.71°

Table 2: Comparison results of proportion of the valid DPD bin

Method	Reverberation time T_{60}		
	0.2s	0.5s	0.8s
WDO	45.3%/39.1%	21.2%/20.3%	14.4%/14.9%
Proposed	92.1%/91.4%	52.8%/47.5%	30.3%/27.5%

Gaussian white noise for a more realistic simulation. A total of 1000 snapshots are generated for analysis, with the first 500 cases overlapped by three plane waves, and the remaining 500 cases dominated by a single source. As shown in Figure 1, in the case of $N = 1$, the detection threshold of both the two types of TF bins are close to one, causing it hard to distinguish them. By contrast, the use of higher SH order ($N = 2, 3$) increases the disparity between DPD and non-DPD bins due to enhanced spatial resolution, suggesting that higher-order RHC vector are more suitable for detecting direct-path components.

Furthermore, we present the results for the proportion of valid DPD bins among the total TF bins. A set of valid DPD bins, $\mathcal{O}_{\text{VAL}}$, is defined as the set of bins where the absolute difference between the estimated azimuth and elevation and the ground-truth is less than 15° , and the proportion is calculated as $\text{Num}(\mathcal{O}_{\text{VAL}})/\text{Num}(\mathcal{O}_{\text{DPD}})$. We compare our approach with the WDO method in both two- and three-source scenarios at various reverberation levels, whereby the corresponding sources' DOA are set as $\Phi_1 = (70^\circ, 100^\circ)$, $\Phi_2 = (130^\circ, 50^\circ)$ and $\Phi_3 = (190^\circ, 100^\circ)$, as shown in Table 2. One can see that the proportion of valid bins decreases for both methods as the reverberation level increases. Compared to the WDO method, however, the proposed approach significantly increases the proportion of valid bins in all cases, highlighting the effectiveness of our DPD component detection.

Table 1 presents the MAEE results of the five methods under varying reverberation levels in both two- and three-source scenarios, whereby each result is averaged over fifty repeated experiments. As expected, the performance of all approaches degrade by the increase of T_{60} , primarily due to the diminishing of direct-path components. Specifically, the WDO method performs the worst due to not using DPD bin detection, thus the smearing effect caused by reverberation severely affects its lo-

calization accuracy. By contrast, the other four methods localize the sources more robustly. The Closed-form method, limited to the first-order SH coefficients, exhibits a lower DPD bin detection capability. The Decoupled method only uses the magnitude information in higher-order RHC vector, which improves its performance but still renders it suboptimal. The DPD-MUSIC method is hard to distinguish the DPD and non-DPD bins under highly reverberation conditions. It is noted that our approach outperforms the reference methods in most scenarios due to reliable and accurate identification of DPD bins. This highlights the strength of the proposed method in enhancing localization accuracy under adverse acoustic environments.

4.2. Real-world Environments

We finally evaluate the proposed algorithm using real-world measurements recorded with a commercial Eigenmike [32] in a reverberant chamber. The room dimensions are $[7, 6, 4]\text{m}$, with $T_{60} \approx 350$ ms and background noise originating from outdoor activity. Three loudspeakers are positioned 1m away from the array at approximate directions of $\Phi_1 = (229^\circ, 98^\circ)$, $\Phi_2 = (314^\circ, 104^\circ)$ and $\Phi_3 = (59^\circ, 92^\circ)$, respectively. All other experimental settings are consistent with those used in the simulations above, and five repetitive trials are conducted for credible results. The MAEE results of five methods (Proposed, DPD-MUSIC, Decoupled, Closed-form and WDO) are $\{3.17^\circ, 3.45^\circ, 3.35^\circ, 3.66^\circ, 6.4^\circ\}$. We note that the localization accuracy is lower than the simulated results due to unavoidable errors in real acoustic scenarios, our method still outperforms the others, which again validate the effectiveness of the proposed algorithm in practical scenarios.

5. Conclusions

In this paper, we address the challenging problem of multi-source DOA estimation in reverberant environments using a spherical array. We propose to apply an OPM to the RHC vector for identifying direct-path components in the STFT domain. Additionally, the determination of OPM is associated with a closed-form DOA estimator, thereby avoiding the tedious search on a spatial grid. Both the simulation and experimental results validate the effectiveness of the proposed algorithm.

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7. References

- [1] T. Latif, E. Whitmire, T. Novak, and A. Bozkurt, "Sound localization sensors for search and rescue biobots," *IEEE Sensors Journal*, vol. 16, no. 10, pp. 3444–3453, 2015.
- [2] A. Jukić, T. van Waterschoot, T. Gerkmann, and S. Doclo, "Multi-channel linear prediction-based speech dereverberation with sparse priors," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 23, no. 9, pp. 1509–1520, 2015.
- [3] C. Pan and J. Chen, "A framework of directional-gain beamforming and a white-noise-gain-controlled solution," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 2875–2887, 2022.
- [4] J. Ahrens, H. Helmholz, D. L. Alon, and S. V. Amengual Garí, "Spherical harmonic decomposition of a sound field based on observations along the equator of a rigid spherical scatterer," *The Journal of the Acoustical Society of America*, vol. 150, no. 2, pp. 805–815, 2021.
- [5] D. P. Jarrett, E. A. Habets, and P. A. Naylor, *Theory and applications of spherical microphone array processing*. Springer, 2017, vol. 9.
- [6] H. Sun, E. Mabande, K. Kowalczyk, and W. Kellermann, "Joint doa and tdoa estimation for 3d localization of reflective surfaces using eigenbeam mvdr and spherical microphone arrays," in *2011 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2011, pp. 113–116.
- [7] Y. Hu, T. D. Abhayapala, and P. N. Samarasinghe, "Multiple source direction of arrival estimations using relative sound pressure based MUSIC," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 253–264, 2020.
- [8] Q. Huang, L. Zhang, and Y. Fang, "Two-stage decoupled doa estimation based on real spherical harmonics for spherical arrays," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 25, no. 11, pp. 2045–2058, 2017.
- [9] D. Pavlidis, S. Delikaris-Manias, V. Pulkki, and A. Mouchtaris, "3D localization of multiple sound sources with intensity vector estimates in single source zones," in *Proc. IEEE 23rd Eur. Signal Process. Conf.*, 2015, pp. 1556–1560.
- [10] S. Hafezi, A. H. Moore, and P. A. Naylor, "Multiple source localization in the spherical harmonic domain using augmented intensity vectors based on grid search," in *2016 24th European Signal Processing Conference (EUSIPCO)*, 2016, pp. 602–606.
- [11] Y. Hu and S. Gannot, "Closed-form single source direction-of-arrival estimator using first-order relative harmonic coefficients," in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 726–730.
- [12] Y. Hu, P. N. Samarasinghe, T. D. Abhayapala, and S. Gannot, "Semi-supervised multiple source localization using relative harmonic coefficients under noisy and reverberant environments," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 28, pp. 3108–3123, 2020.
- [13] O. Nadiri and B. Rafaely, "Localization of multiple speakers under high reverberation using a spherical microphone array and the direct-path dominance test," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 22, no. 10, pp. 1494–1505, 2014.
- [14] A. Brendel, C. Huang, and W. Kellermann, "Stft bin selection for localization algorithms based on the sparsity of speech signal spectra," *ratio*, vol. 2, p. 6, 2018.
- [15] H. Wang, K. Chen, and J. Lu, "U-net based direct-path dominance test for robust direction-of-arrival estimation," *arXiv preprint arXiv:2005.04376*, 2020.
- [16] S. Mohan, M. E. Lockwood, M. L. Kramer, and D. L. Jones, "Localization of multiple acoustic sources with small arrays using a coherence test," *The Journal of the Acoustical Society of America*, vol. 123, no. 4, pp. 2136–2147, 2008.
- [17] B. Rafaely and K. Alhaiani, "Speaker localization using direct path dominance test based on sound field directivity," *Signal Processing*, vol. 143, pp. 42–47, 2018.
- [18] D. Pavlidis, A. Griffin, M. Puigt, and A. Mouchtaris, "Real-time multiple sound source localization and counting using a circular microphone array," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 21, no. 10, pp. 2193–2206, 2013.
- [19] Y. Hu, P. N. Samarasinghe, T. D. Abhayapala, and S. Gannot, "Unsupervised multiple source localization using relative harmonic coefficients," in *2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 571–575.
- [20] Y. Hu, P. N. Samarasinghe, S. Gannot, and T. D. Abhayapala, "Decoupled multiple speaker direction-of-arrival estimator under reverberant environments," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 3120–3133, 2022.
- [21] Y. Hu, W. Wang, Z. Gu, T. Mao, X. Zhu, and J. Jin, "Closed-form multiple source direction-of-arrival estimator under reverberant environments," *The Journal of the Acoustical Society of America*, vol. 154, no. 4, pp. 2349–2364, 2023.
- [22] P. N. Samarasinghe, T. D. Abhayapala, and H. Chen, "Estimating the direct-to-reverberant energy ratio using a spherical harmonics-based spatial correlation model," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 25, no. 2, pp. 310–319, 2016.
- [23] B. Bu, C. Bao, and M. Jia, "Design of a planar first-order loudspeaker array for global active noise control," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 26, no. 11, pp. 2240–2250, 2018.
- [24] D. B. Ward and T. D. Abhayapala, "Reproduction of a plane-wave sound field using an array of loudspeakers," *IEEE Transactions on speech and audio processing*, vol. 9, no. 6, pp. 697–707, 2001.
- [25] I. Balmages and B. Rafaely, "Open-sphere designs for spherical microphone arrays," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 15, no. 2, pp. 727–732, 2007.
- [26] S. Gao, J. Lin, X. Wu, and T. Qu, "Sparse dnn model for frequency expanding of higher order ambisonics encoding process," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 30, pp. 1124–1135, 2022.
- [27] L. Tao, M. Jia, C. Bao, and W. Xiong, "First-order relative harmonic coefficient-based time-frequency points selection for multi-source doa estimation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2024.
- [28] Y.-C. Chen, "A tutorial on kernel density estimation and recent advances," *Biostatistics & Epidemiology*, vol. 1, no. 1, pp. 161–187, 2017.
- [29] J. B. Allen and D. A. Berkley, "Image method for efficiently simulating small-room acoustics," *J. Acoust. Soc. Amer.*, vol. 65, no. 4, pp. 943–950, 1979.
- [30] V. Zue, S. Seneff, and J. Glass, "Speech database development at mit: Timit and beyond," *Speech communication*, vol. 9, no. 4, pp. 351–356, 1990.
- [31] A. Fahim, P. N. Samarasinghe, and T. D. Abhayapala, "Psd estimation and source separation in a noisy reverberant environment using a spherical microphone array," *IEEE/ACM Transactions on audio, speech, and language processing*, vol. 26, no. 9, pp. 1594–1607, 2018.
- [32] E. Pfanzagl-Cardone, "HOA—higher order ambisonics (eigenmike®)," in *The Art and Science of 3D Audio Recording*. Springer, 2023, pp. 189–209.